Online Indoor Visual Odometry with Semantic Assistance under Implicit Epipolar Constraints

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***Abstract*—Indoor visual SLAM (Simultaneous Localization And Mapping) has been widely used in the tasks of estimating the position of the carrier and reconstructing the indoor structure with fine details. In these tasks, compared with traditional SLAM, learning-based VO (Visual Odometry) has gained more and more popularity due to its robustness and low cost. However, the performance of existing indoor deep VO schemes is still limited in comparison with their outdoor counterparts mainly owning to the following two reasons: *i*) large areas of textureless regions can be more frequently seen indoors such as walls, blackboards, *etc*, which contain extremely sparse photometric features; *ii*) the complex indoor motions cause more difficulty to the pose regression network than the outdoor motions which are dominated by simple translation components. In this paper, the above two challenges are carefully tackled with the proposed SEOVO (Semantic Epipolar-constrained Online VO), introducing both the semantic and the geometric cues in an online manner to improve the performance of the depth and the pose estimation. On the one hand, as far as we know, SEOVO is the first semantic-aided VO under an online adaptive framework, which adaptively handles the low-texture planes without any supervision, guaranteeing the reconstruction results in these challenging areas with better generalization ability. Specifically, the textureless planes are nicely extracted and segmented by a "coarse-to-fine" strategy with the help of the semantic segmentation results and our designed "multi-grad" map. On the other hand, in existing solutions, the camera pose is learned rather than calculated analytically, thus limiting the regression performance of the PoseNet. To relieve this problem to some extent, we introduce the epipolar geometric constraint in an implicit way for improving the accuracy of the pose estimation without destroying the global scale consistency. In addition, as mentioned before, a "multi-grad" map is designed to fuse the multi-source gradients induced by the photometric, the semantic and the geometric constraints, which is of great use not only in the fine extraction of textureless planes, but also in the generation of a sharp depth map over the edges. The efficiency and efficacy of SEOVO have been corroborated by extensive experiments conducted on both public datasets and our collected video sequences. To make our results reproducible, our codes and data are available at https://cslinzhang.github.io/SEOVO/SEOVO.html.**

***Index Terms*—Indoor visual odometry, self-supervised learning, unsupervised semantic segmentation, epipolar geometric constraint, online learning.**

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I. Introduction

WITH the development of various 3D visual perceptual tasks such as smart homes, 3D indoor navigation and Augmented Reality (AR), accurate localization and high-quality 3D indoor reconstruction are two basic tasks which attract significant attention. Fortunately, recent years, visual SLAM has been developed rapidly as an excellent solution to both tasks, and is successfully applied to various mobile reconstruction platforms, such as the mobile robot, the mobile backpack and the micro UAV (Unmanned Aerial Vehicle).

As an indispensable part in visual SLAM, VO (Visual Odometry) is responsible for calculating the camera pose between adjacent frames and recovering the local map. According to the optimization frameworks adopted, existing VO schemes mainly fall into two categories, the traditional ones [1]–[7], [9]–[12] and the learning-based ones [13]–[28], and their limitations are mainly summarized as follows:

1) The traditional VO schemes suffer from low-level manual features, which are fragile to the illumination variations and the texture distributions. Nearly all of the classic SLAM systems [1]–[6] are built on the golden rule of feature correspondence and often perform reliably in ideal environments with rich textures. However, the robustness of these point-based SLAM systems in low-texture regions needs to be improved. Later, line-based SLAMs [7], [8] and plane-based SLAMs [9]–[12] came into existence. However, the former ones rely on the line descriptors which are time-consuming in extraction while the latter ones usually assume that the camera poses are known. Also, their utilized line (plane) features are still low-level and handmade, leading to that the extractions of lines and planes are vulnerable to occlusion and the damage of edge patterns.
2) Existing learning-based VO schemes usually have limitations in generalization abilities whether they are supervised or not. Compared with traditional VOs, they extract high-level features through convolutional networks and jointly learn the prediction of monocular depth and ego-motion, showing excellent performance in locating and mapping. However, the majority of them [13]–[21],

[24]–[28] are based on offline learning which optimize network models on the training data and inference on the testing data with model parameters fixed. Thus, the performance of these offline VOs will significantly decrease when working in new scenes different from the training set. In addition, few learning-based VO methods pay attention to the challenging indoor environments which bring difficulty to the regression of depths and poses due to the low-texture regions and complex indoor motions. Thus, the indoor performance of learning-based VOs is still limited.

On account of the aforementioned limitations, we aim to deal with the indoor challenges by introducing both the semantic and the geometric information of indoor scenes and improve the generalization ability of the learning framework through online optimization. Specifically, we propose an online adaptive VO with semantic assistance under an implicit epipolar constraint, namely SEOVO (Semantic Epipolar-constrained Online VO). To our best knowledge, SEOVO is the first online monocular deep VO with semantic assistance. In summary, our contributions are mainly threefold:

1) The first semantics-aided VO under an online adaptive framework, SEOVO, is proposed. Different from offline training schemes which inference on the testing data with the model parameters fixed, SEOVO is optimized under an online meta-learning framework [22], adapting itself to every new frame. Besides, although there exist several unsupervised methods leveraging semantic information [28]–[30] for monocular depth estimation, they either rely on the pretrained semantic segmentation network with fixed weights or need manually labelled class labels for multi-task training. By contrast, our SEOVO can adapt the model online and does not require any semantic inputs, which is of great help in adaptively coping with the textureless regions.
2) A novel epipolar constrained photometric loss is designed to implicitly introduce the epipolar geometric constraint free of the scale uncertainty problem to guide the pose regression. The optimization of most existing self-supervised VOs is dominated by the traditional photometric loss and ignores the geometric relationships in the scene, limiting the system performance especially for the pose estimation. By contrast, SEOVO introduces the epipolar geometric constraint of pixel correspondences and determines the scale of the epipolar pose which is solved from the fundamental matrix by aligning it with the predicted pose. Based on the scaled pose, our newly designed photometric loss can be formed to impose the two-view geometric constraint on the pose regression without ruining the global scale consistency compared with explicitly taking the epipolar pose as direct supervision. What's more, we also conduct the global optimization of the reconstructed map to relieve the accumulation draft, further improving the accuracy of the pose estimation.
3) A "multi-grad" map fusing the gradients embedded with the photometric, the semantic and the geometric information is designed. Compared with the single photometric gradient map, this fused gradient map can better distinguish different instances especially in textureless regions. Additionally, the "multi-grad" map can be of great benefit in ensuring a sharp depth map along the object edges. To sum up, it encourages SEOVO to give predictions consistent with the photometric, the semantic and the geometric constraints.

The remainder of this paper is organized as follows. Sect. II introduces related studies. Sect. III provides some preliminary knowledge for better understanding this article. Details of the proposed SEOVO are presented in Sect. IV. Experimental results are reported in Sect. V. Finally, Sect. VI concludes the paper.

II. Related Work

A. Self-supervised Depth and Pose Estimation

Early works [31]–[38] of depth estimation are mainly supervised and achieve excellent performance, but it is expensive to capture ground-truth data in many real-world scenes. Compared with the above schemes, jointly training the depth and pose network from unlabelled monocular videos shows its simplicity and effectiveness, which attracts a lot of researchers' interests and inspires a series of works [13]–[28] including ours. According to the target environments, existing self-supervised VO methods predicting from monocular videos are mainly divided into two categories, the outdoor ones and the indoor ones.

Outdoor self-supervised VO. Zhou *et al.* [13] were the first to jointly regress the depth and the pose using a self-supervised framework based on the photometric loss across temporal frames. Many follow-up methods then sprang up to improve the performance. Godard *et al.* [14] proposed a novel left-right disparity consistency loss term mocking the stereo image pairs. Hence, their training process required rectified and temporally aligned stereo pairs. SC-SfMLearner [15], [16] emerged as a public baseline work, aiming to solve two crucial problems in monocular VO. First, they put forth a scale consistency loss to restrict the depth consistency between adjacent frames, thus easing the scale ambiguity problem and building a globally consistent map. Second, based on the first point, they designed a mask to remove the dynamic or occluded regions in the scene. Wang *et al.* [19] and Zou *et al.* [20] leveraged recurrent neural networks, such as LSTMs, to dig the long-term connection of the poses and/or the depths among several consecutive frames. Monodepth2 [21] significantly improved the performance over previous methods in the aspects of occlusion handling. Nonetheless, it masked out pixels that are relatively static to the camera, which discourages the pose prediction. To obtain a better generalization capability, online learning was introduced for online update of the pose and depth estimation in [22] and [23], respectively.

Indoor self-supervised VO. The motions involved in outdoor datasets are dominated by translations, which are easier for the pose network to regress. Lately, researchers got to find that current approaches could not achieve comparative performance on indoor datasets as outdoor ones. To fill in this gap to

some extent, a few solutions focusing on the characteristics of indoor environments and indoor motions emerged. Bian *et al.* [24] considered the rotation as the "noise during training" and imitated the stereo rectification to remove the adjacent frames with nearly pure rotation. Although it works well on their rectified dataset, the removing strategy cannot produce plausible results in unseen environments since no rotation motion is hard to be promised especially in indoor data. MonoIndoor [25] took a different strategy of progressively estimating the rotations via a novel residual pose module instead of directly removing them. MovingIndoor [26] proposed a new optical-flow based training paradigm which reduced the difficulty of unsupervised learning by utilizing the results of pretrained optical-flow network as the supervision. Zhao *et al.* [27] employed a differentiable two-view triangulation layer to generate a sparse depth map as the self-supervision on depth. However, its triangulated depth is sensitive to mismatch. Notably, few of the aforementioned works explicitly deal with the challenging weakly textured scenes, which are commonly seen in indoor environments and have not been well coped with yet.

B. Solutions to Low-Texture Environments

Most traditional visual SLAM systems are based on point features [1]–[6]. However, in some low-texture environments, these features are extremely sparse or distributed unevenly, leading to the degradation or even failure of the system. Therefore, the line-based [**?**] and the plane-based [9]–[12] SLAM systems were developed. Yang *et al.* [**?**] extracted sufficient line features on the premise of ensuring real-time performance, so as to obtain a visual SLAM with higher accuracy and robustness. Planes [9], [11] or super-pixels [10] were also used to reconstruct the low-texture areas. However, these plane-based methods assumed that the pose was provided by traditional SLAM and only focused on the depth estimation. Pop-up SLAM [12] popped the plane based on the room layout and fused point-based and plane-based SLAM altogether to enhance depth estimation. The aforementioned methods are all based on low-level features and tend to fail when the target regions are occluded or the boundaries are broken. Instead, Zhou *et al.* [26] resorted to deep learning to capture high-level features in the scene. In their work, a well-designed SF-Net actively propagated the sparse initial flows at low-textured regions to the entire image. Although it improves the performance in low-textured regions in most cases, it requires a certain amount of sparse seeds which cannot be satisfied when the low-textured regions are relatively large. On the other side, several approaches resorted to semantic understanding for better indoor reconstruction. For example, Concha *et al.* [39] suggested using the layout of the room to generate prior depth map for dense mapping, but they didn't track and update the layout, so their solution can only work in small space.

III. Preliminaries

In this section, we provide some preliminaries for better understanding our work. First, we give a brief introduction of the common loss terms widely utilized in self-supervised VOs, learning from which we design our optimization objectives. Second, we briefly recap the framework of online meta-learning to help understand its effectiveness in improving the generalization ability of the learning model.

A. Common Loss Terms of Self-supervised VO

Photometric loss. The photometric loss appears in the loss functions of nearly all the self-supervised works by penalizing the difference between the target frame and the warped one synthesized by the predicted pose and depth. Specifically, with the target image $\boldsymbol{I}_t$ and the source image $\boldsymbol{I}_s$ as inputs, the network can output the local transformation matrix from the target view to the source one denoted by $\boldsymbol{T}^L_{t\to s}$ and their individual depth maps denoted by $\boldsymbol{D}_t$ and $\boldsymbol{D}_s$, respectively. The warping transformation can be formulated as:

$$\boldsymbol{p}'_t = \frac{1}{Z_{\boldsymbol{p}'_t}} \boldsymbol{K}\boldsymbol{T}^L_{t\to s}\boldsymbol{D}_t(\boldsymbol{p}_t)\boldsymbol{K}^{-1}\boldsymbol{p}_t \tag{1}$$

where $\boldsymbol{K}$ denotes the camera intrinsic matrix, $Z_{\boldsymbol{p}'_t}$ is the depth of the warped 3D point in the adjacent camera's coordinate system, and $\boldsymbol{p}_t$ and $\boldsymbol{p}'_t$ are the pixel coordinates before and after warping. Based on the mapping relationship in Eq. 1, we can sample in $\boldsymbol{I}_s$ to obtain the warped image $\boldsymbol{I}'_t$ by

$$\begin{aligned}\boldsymbol{I}'_t(\boldsymbol{p}_t) &= \boldsymbol{I}_s(\boldsymbol{p}'_t)\\ &= \boldsymbol{I}_s(\frac{1}{Z_{\boldsymbol{p}'_t}}\boldsymbol{K}\boldsymbol{T}^L_{t\to s}\boldsymbol{D}_t(\boldsymbol{p}_t)\boldsymbol{K}^{-1}\boldsymbol{p}_t)\end{aligned} \tag{2}$$

Based on Eq. 2, the photometric loss calculates the distance between $\boldsymbol{I}_t$ and $\boldsymbol{I}'_t$ as [16]:

$$\begin{aligned}L_{pho} &= \frac{1}{\|\boldsymbol{\mathcal{V}}\|}\sum_{\boldsymbol{p}_t\in\boldsymbol{\mathcal{V}}}[\lambda\left\|\boldsymbol{I}_t(\boldsymbol{p}_t)-\boldsymbol{I}'_t(\boldsymbol{p}_t)\right\|_1 + (1-\lambda)\frac{1-SSIM_t(\boldsymbol{p}_t)}{2}]\\ &= \frac{1}{\|\boldsymbol{\mathcal{V}}\|}\sum_{\boldsymbol{p}_t\in\boldsymbol{\mathcal{V}}}[\lambda\left\|\boldsymbol{I}_t(\boldsymbol{p}_t)-\boldsymbol{I}_s(\frac{1}{Z_{\boldsymbol{p}'_t}}\boldsymbol{K}\boldsymbol{T}^L_{t\to s}\boldsymbol{D}_t(\boldsymbol{p}_t)\boldsymbol{K}^{-1}\boldsymbol{p}_t)\right\|_1\\ &\quad + (1-\lambda)\frac{1-SSIM_t(\boldsymbol{p}_t)}{2}]\end{aligned} \tag{3}$$

where $\boldsymbol{\mathcal{V}}$ stands for the set of valid points without boundary or occluded points. $SSIM_t(\cdot)$ calculates element-wise similarity between $\boldsymbol{I}_t$ and $\boldsymbol{I}'_t$ by the Structured Similarity (SSIM) function [40], and $\lambda = 0.15$.

Depth smoothness loss. A depth smoothness loss L_{sm} is employed to regularize the depth estimation, which indeed ensures the smoothness of the depth map to be guided by the image gradient. Formally,

$$L_{sm} = \frac{1}{\|\boldsymbol{\mathcal{V}}\|}\sum_{\boldsymbol{p}_t\in\boldsymbol{\mathcal{V}}}(exp(-\boldsymbol{G}_{pho}(\boldsymbol{p}_t))\cdot\nabla\boldsymbol{D}_t(\boldsymbol{p}_t))^2 \tag{4}$$

where ∇ is the gradient operator and $\boldsymbol{G}_{pho}$ is the gradient map of the target image calculated by $\boldsymbol{G}_{pho} = \nabla\boldsymbol{I}_t$.

Scale consistency loss. The scale consistency loss L_{sc} constrains the frame-to-frame depth scale to maintain the globally consistent map. Given the source depth map $\boldsymbol{D}_s$ and the synthesized one $\boldsymbol{D}'_s$ generated by the target depth map $\boldsymbol{D}'_t$ and the pose $\boldsymbol{T}^L_{t\to s}$, the scale consistency loss is defined as:

$$L_{sc} = \frac{1}{\|\boldsymbol{\mathcal{V}}\|}\sum_{\boldsymbol{p}_t\in\boldsymbol{\mathcal{V}},}\frac{\left\|\boldsymbol{D}_s(\boldsymbol{p}_t)-\boldsymbol{D}'_s(\boldsymbol{p}_t)\right\|}{\boldsymbol{D}_s(\boldsymbol{p}_t)+\boldsymbol{D}'_s(\boldsymbol{p}_t)}. \tag{5}$$

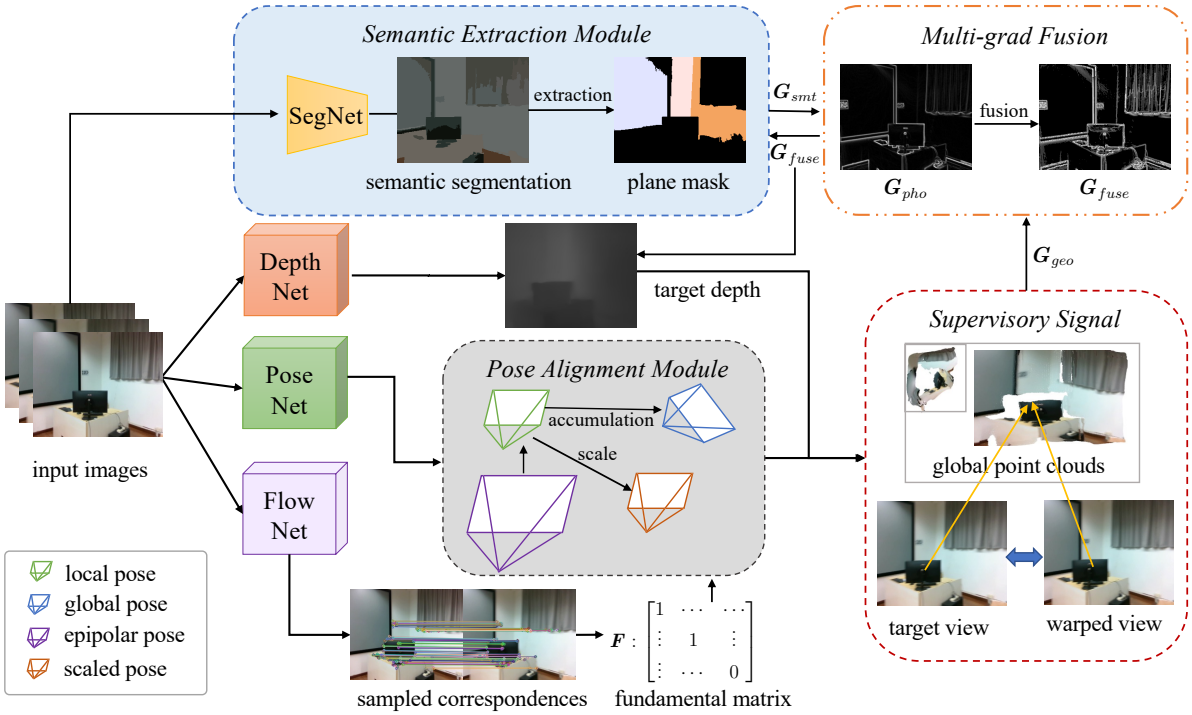

Fig. 1. The system pipeline of our SEOVO. The DepthNet and the PoseNet are used to estimate the depth map and the relative pose, respectively. The FlowNet predicts the optical flow between adjacent frames to provide the epipolar pose. Two innovative modules are proposed. The "Pose Alignment Module" introduces the geometric constraints to the pose regression through the alignments of the predicted pose while the "Semantic Extraction Module" utilizes the semantic prior to extract the low-texture planes for constraining them with planar constraints. Also, we generate a multi-grad map $\boldsymbol{G}_{fuse}$ through "Multi-grad Fusion" to fuse the photometric, the semantic and the geometric gradients denoted by $\boldsymbol{G}_{pho}, \boldsymbol{G}_{smt}$ and $\boldsymbol{G}_{geo}$, respectively. The supervisory signals are mainly composed of the photometric and the geometric consistency between frames.

B. Online Meta-learning

Most learning algorithms assume that the training and testing data are sampled from the same feature distribution, which cannot be promised in all unseen environments, thus profoundly restricting the performance of the pretrained models. Therefore, we require the model to adapt to the new data online, which means it still needs to learn in the testing phase. Reviewing the optimization objective in conventional online learning, given a model parameterized by $\boldsymbol{\theta}$ pretrained on training dataset $\mathcal{D}^{train}$, a naive approach for online updating $\boldsymbol{\theta}$ is to back-propagate the loss L on the current sample $\boldsymbol{s}_i$:

$$\boldsymbol{\theta}_{i+1} = \boldsymbol{\theta}_i - \alpha \nabla L(\theta_i, \boldsymbol{s}_i) \tag{6}$$

where α is the learning rate. Inspired by meta-learning, Li *et al.* [22] extended the learning objective with a larger temporal perceptive field, which can be written as:

$$\min_{\boldsymbol{\theta}_i} L(\boldsymbol{\theta}_i - \alpha \nabla L(\boldsymbol{\theta}_i, \boldsymbol{s}_i), \boldsymbol{s}_{i+1}) \tag{7}$$

Different from naive online learning, in Eq. 7, the optimization variable with respect to the loss on $\boldsymbol{s}_{i+1}$ is $\boldsymbol{\theta}_i$ instead of $\boldsymbol{\theta}_{i+1}$, compelling the network to learn the best $\boldsymbol{\theta}_i$ which can still perform well at next time $i+1$.

IV. SEOVO

The overall pipeline of our SEOVO is shown in Fig. 1. It takes consecutive frames from the monocular video as inputs and utilizes three main networks, the DepthNet, the PoseNet, and the FlowNet to predict the depth of the target image, the relative pose between adjacent frames and the optical flow, respectively. Lying at the core of the SEOVO are two innovative modules, namely the "Pose Alignment Module" and the "Semantic Extraction Module". Specifically, in the pose alignment module, a scaled pose and a global pose are obtained through two alignment transformations, respectively. The scaled pose implicitly introduces the epipolar constraint while the global one helps to alleviate the accumulation drift problem. Besides, in the semantic extraction module, we utilize an unsupervised semantic segmentation network denoted by "SegNet" to provide the semantic prior, based on which textureless planes can then be nicely extracted and segmented. Another important innovation of SEOVO is that a multi-grad map is obtained through the multi-grad fusion to integrate the gradients of the input image, the segmentation map and the synthesized coarse point clouds. The edge information provided by this gradient map not only better guides the optimization of the depth map, but also helps to generate a finer extraction result of low-texture planes. In the following, we will give detailed explanations to the key components of SEOVO, the pose alignment module, the semantic extraction module, the multi-grad fusion and the loss function, respectively.

A. Pose Alignment Module

The pose alignment module is responsible for generating two aligned poses to introduce geometric constraints into

SEOVO, which are the scaled pose $\boldsymbol{T}^S$(the subscript $t \rightarrow s$ is omitted for brevity) and the global pose $\boldsymbol{T}^G$. Denoting the local predicted pose from PoseNet by $\boldsymbol{T}^L$, and the epipolar pose solved from the fundamental matrix based on the optical flow from FlowNet by $\boldsymbol{T}^E$, this module conducts two alignments to obtain $\boldsymbol{T}^S$ and $\boldsymbol{T}^G$, respectively, as illustrated in Fig. 2. On the one hand, $\boldsymbol{T}^S$ is generated by aligning $\boldsymbol{T}^E$ with $\boldsymbol{T}^L$ in scale, namely scale alignment. On the other hand, $\boldsymbol{T}^L$ is transformed to the global coordinate system to obtain $\boldsymbol{T}^G$, namely local-global alignment. Next we will specify these two transformations in detail.

Scale alignment. As we know, indoor sequences contain much more rotational motions than outdoor ones, bringing difficulty to poseNet in the pose regression. We try to solve this problem by introducing the epipolar geometric constraint. However, owning to the fact that the translation vector of the epipolar pose $\boldsymbol{T}^E$ is normalized, explicitly utilizing $\boldsymbol{T}^E$ as the direct supervision of PoseNet will suffer from the scale ambiguity problem. One common solution is to normalize the predicted pose $\boldsymbol{T}^L$ so as to penalize the distance between it and $\boldsymbol{T}^E$, which is considered as an explicit way to introduce the epipolar geometric constraint. Unfortunately, we find that this explicit strategy can easily damage the global scale consistency since the normalization operation discards the scale of the predicted translation vector, which determines the scale of the whole map. Instead, we align $\boldsymbol{T}^E$ with the predicted one $\boldsymbol{T}^L$ to produce the scaled pose $\boldsymbol{T}^S$ as:

$$\boldsymbol{T}^S = \begin{bmatrix} \boldsymbol{R}^E & s \cdot \boldsymbol{t}^E \\ \boldsymbol{0}^T & 1 \end{bmatrix}, s = \frac{\boldsymbol{t}^L[i]}{\boldsymbol{t}^E[i]}, i = \underset{0 \leq i \leq 2}{\arg\max} |\boldsymbol{t}^L[i]| \quad (8)$$

where $\boldsymbol{R}^E, \boldsymbol{t}^E$ are the rotation matrix and the translation vector of $\boldsymbol{T}^E$ respectively, and $\boldsymbol{t}^L$ is the translation vector in $\boldsymbol{T}^L$. $\boldsymbol{t}[i]$ returns the i-th element in the translation vector $\boldsymbol{t}$. Through Eq. 8, $\boldsymbol{T}^S$ keeps the reliable rotation matrix and translational direction calculated based on the epipolar constraint, while preserving the scale of the predict pose at the same time.

Local-global alignment. Accumulation drift is a common problem in the long-term localization of the visual SLAM system. This is because the visual odometry only estimates the motion between two frames and each prediction inevitably has a certain error, which will be transferred to the estimation at the next time. These accumulated errors lead the estimated trajectory to be inaccurate after a period of time. To deal with the problem, we propose to conduct the global optimization by keeping the consistency of the global map. Therefore, we need to prepare the global pose of the current frame taking the first frame as the reference. In detail, with the local pose $\boldsymbol{T}^L_{(k-1)\rightarrow k}$ at time k, we transform it to the global coordinate system to obtain the global pose $\boldsymbol{T}^G_k$ by multiplying all the pose estimations over the past time. Specifically, $\boldsymbol{T}^G_k$ can be formulated as:

$$\boldsymbol{T}^G_k = \prod_{i=1}^{k} \boldsymbol{T}^L_{(i-1)\rightarrow i}. \quad (9)$$

So far, we have obtained the target $\boldsymbol{T}^S$ and $\boldsymbol{T}^G$. Then the corresponding loss terms L'_{pho} and L_{pc} can be generated, which are given in Eq. 14 and Eq. 16, respectively. By integrating these two loss terms in the loss function of SEOVO, we can ensure the multi-view geometric consistency in the pose regression.

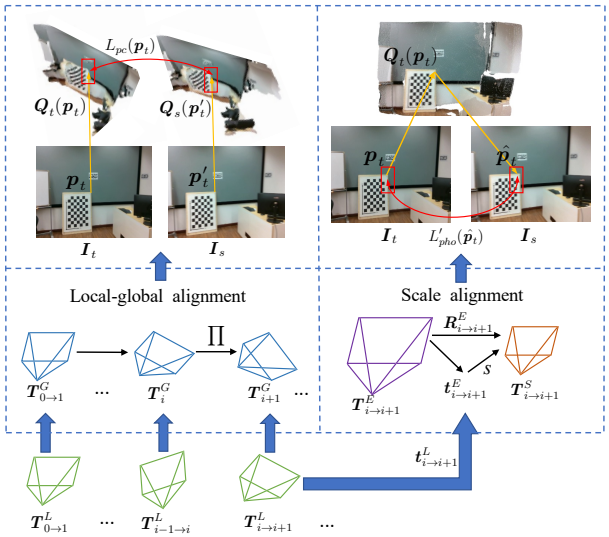

Fig. 2. Illustration of two alignment steps in the "Pose Alignment Module" and the corresponding loss terms, including the point cloud loss L_{pc} based on the global pose $\boldsymbol{T}^G$ and the epipolar constrained photometric loss L'_{pho} utilizing the scaled pose $\boldsymbol{T}^S$. The former one maintains the global consistency of the synthesized point clouds while the latter one introduces the epipolar constraint without scale ambiguity.

B. Semantic Extraction Module

Considering that addressing the low-texture environment is a challenging problem in indoor VOs, the semantic extraction module is designed to extract the textureless planes so that planar constraints can be imposed on these regions to ensure the quality of reconstruction. Technically, we propose a coarse-to-fine extraction strategy containing two stages, namely the coarse extraction and the fine extraction. The whole extraction algorithm is given in Alg. 1.

Coarse extraction. In order to extract the textureless regions in the target image $\boldsymbol{I}_t$, we design a screening principle based on the "photometric overfitting" phenomenon. Specifically, we have an interesting insight that the photometric errors of the pixel correspondences tend to be extremely small in the textureless areas. The underlying reason is that pixels in these areas usually have similar colors, and small photometric errors can be kept even if the correspondences are mismatched, which is called the phenomenon of "photometric overfitting". To make use of this phenomenon to help extract textureless planes, we first form an photometric error map $\boldsymbol{M}_{pho}$ where $\boldsymbol{M}_{pho}(\boldsymbol{p}_t)$ records the photometric error of an arbitrary pixel $\boldsymbol{p}_t$ in $\boldsymbol{I}_t$. Then 2D points with small photometric errors are extracted through Eq. 10 to form a coarse textureless pixel set $\boldsymbol{\mathcal{C}}$.

$$\boldsymbol{\mathcal{C}} = \{\boldsymbol{p}_i | \boldsymbol{M}_{pho}(\boldsymbol{p}_i) < \mu_{pho} + \sigma_{pho}, \boldsymbol{p}_i \in \boldsymbol{I}_t\} \quad (10)$$

where μ_{pho} and σ_{pho} are the mean and the variance of $\boldsymbol{M}_{pho}$, respectively. Having obtained the sparse textureless points, we gather the corresponding class labels of all $\boldsymbol{p}_i$ in set $\boldsymbol{\mathcal{L}}$ based on the segmentation map $\boldsymbol{M}_{seg}$ predicted by SegNet. Assuming

that the pixels of the same class in the textureless regions are on the same plane, we collect pixels whose semantic labels are in $\boldsymbol{\mathcal{L}}$ and divide them into different planes according to their classes to form the coarse set of textureless planes denoted by $\boldsymbol{\mathcal{P}}^C$.

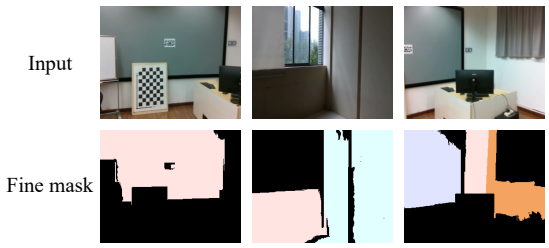

Fig. 3. The visualization results of the final extracted textureless planes in different scenes. Different planes are filled with different colors

Fine extraction. In practice, the extraction strategy in the coarse stage is not accurate enough to separate all different planes because the coarse extraction only focuses on the photometric and semantic information, based on which it's usually difficult to separate planes with the same color and semantics. A typical example is offered in Fig. 4 (b). To deal with this problem, we design the fine stage to further refine the coarse extraction results, including kicking out outliers and obtaining a finer segmentation result of current extracted regions. On the one hand, for each coarse region $P^C \in \boldsymbol{\mathcal{P}}^C$, only their largest connected region P_{lcr} is kept to get rid of small planar regions in consideration of robustness and efficiency. On the other hand, we borrow advantage of the multi-grad map $\boldsymbol{G}_{fuse}$ providing richer edges than the image gradient to separate instances with the same semantic labels, which will be introduced in Sect. IV-C. Technically, for each connected region P_{lcr}, we detect the separating line in the same area of $\boldsymbol{G}_{fuse}$. If the separating line is detected, it divides P_{lcr} into two perpendicular planes denoted by P_1 and P_2 which are grouped and then collected in the set of paired planes $\boldsymbol{\mathcal{P}}^P$. Otherwise, P_{lcr} is added to the set $\boldsymbol{\mathcal{P}}^I$ containing independent planes. The final extraction results are illustrated in Fig. 3.

Algorithm 1 Coarse-to-fine extraction of textureless planes

Input: The target image $\boldsymbol{I}_t$
The multi-grad map $\boldsymbol{G}_{fuse}$
The segmentation map $\boldsymbol{M}_{seg}$
Output: Set of independent planes $\boldsymbol{\mathcal{P}}^I$
Set of paired planes $\boldsymbol{\mathcal{P}}^P$

Calculate the photometric error map $\boldsymbol{M}_{pho}$ of $\boldsymbol{I}_t$
$u_{pho} \leftarrow$ the mean of $\boldsymbol{M}_{pho}$
$\sigma_{pho} \leftarrow$ the variance of $\boldsymbol{M}_{pho}$
$\boldsymbol{\mathcal{P}}^C$: the set of the coarse planes in coarse stage
$\boldsymbol{\mathcal{L}}$: the set of class labels of coarsely extracted pixels
for all $\boldsymbol{p}_t \in \boldsymbol{I}_t$ **do**
  if $\boldsymbol{M}_{pho}(\boldsymbol{p}_t) < \mu_{pho} + \sigma_{pho}$ **then**
    $\boldsymbol{p}_i \leftarrow \boldsymbol{p}_t$
    $l_i \leftarrow \boldsymbol{M}_{seg}(\boldsymbol{p}_i)$
    if $l_i \notin \boldsymbol{\mathcal{L}}$ **then**
      Insert l_i to $\boldsymbol{\mathcal{L}}$
      $P^C \leftarrow \{\boldsymbol{p}_{tpt} | \boldsymbol{M}_{seg}(\boldsymbol{p}_{tpt}) = l_i, \boldsymbol{p}_{tpt} \in \boldsymbol{I}_t\}$
      Insert P^C to $\boldsymbol{\mathcal{P}}^C$
    end if
  end if
end for
for all $P^C \in \boldsymbol{\mathcal{P}}^C$ **do**
  $P_{lcr} \leftarrow$ the largest connected region of P^C
  $\boldsymbol{d} \leftarrow$ detect line in $\boldsymbol{G}_{fuse}(P_{lcr})$
  if $\boldsymbol{d}$ is not None **then**
    $P_1 \leftarrow$ the plane above $\boldsymbol{d}$
    $P_2 \leftarrow$ the plane below $\boldsymbol{d}$
    Insert P_1, P_2 to $\boldsymbol{\mathcal{P}}^I$
    Insert $[P_1, P_2]$ to $\boldsymbol{\mathcal{P}}^P$
  else
    Insert P_{lcr} to $\boldsymbol{\mathcal{P}}^I$
  end if
end for

C. Multi-grad Fusion

The traditional depth smoothness loss L_{sm} in Eq. 4 aims to produce satisfactory depth estimations in textureless areas relying on the image gradient map $\boldsymbol{G}_{pho}$ to guide depth optimization. However, $\boldsymbol{G}_{pho}$ based on the image appearance will ignore the boundaries of objects with similar colors and produce over-smooth depths in these places. For example, as shown in Fig. 4, the wall joint and the desk edge are difficult to be observed from the photometric gradient map $\boldsymbol{G}_{pho}$. Fortunately, we find that the semantic layout and the geometric structure of the scene can compensate for the lack of photometric information in such a situation, thus helping capture more edges consistent with perception. Therefore, on the basis of photometric gradient map $\boldsymbol{G}_{pho}$, we also generate semantic and geometric gradient maps denoted by $\boldsymbol{G}_{smt}$ and $\boldsymbol{G}_{geo}$ respectively to produce the fused gradient map $\boldsymbol{G}_{fuse}$, namely the multi-grad map. Based on $\boldsymbol{G}_{fuse}$, we form a fused smoothness loss term denoted by L'_{sm}, the specific formulation of which will be given in Eq. 20. In addition, the edges detected in $\boldsymbol{G}_{fuse}$ are also used in the fine stage of the semantic extraction module discussed in Sect. IV-B to obtain more accurate segmentation results of textureless planes. Next we will discuss the two newly introduced gradient maps $\boldsymbol{G}_{smt}$ and $\boldsymbol{G}_{geo}$ and how to obtain $\boldsymbol{G}_{fuse}$ in detail.

Semantic gradient. Thanks to the development of the unsupervised semantic segmentation, SegNet can effectively extract high-level features and distinguish objects with different semantics. Therefore, the semantic gradient outperforms the photometric one in discovering edges consistent with the semantic understanding. As shown in Fig. 4 (a), the semantic gradient $\boldsymbol{G}_{smt}$ is able to separate the desk edge from the wall although they have similar colors, which makes them hard to be distinguished by $\boldsymbol{G}_{pho}$. Technically, to obtain $\boldsymbol{G}_{smt}$, we first generate the class label $l(\boldsymbol{p}_t)$ of each pixel $\boldsymbol{p}_t$ by SegNet. For each class $l(\boldsymbol{p}_t)$, we calculate the average density of the pixels belonging to it denoted by $c(l(\boldsymbol{p}_t))$. Then a pseudo colored segmentation map denoted by $\boldsymbol{I}_{smt}$ can be generated by $\boldsymbol{I}_{smt}(\boldsymbol{p}_t) = c(l(\boldsymbol{p}_t))$, and the semantic gradient can be given as $\boldsymbol{G}_{smt} = \nabla \boldsymbol{I}_{smt}$. The visualization results of $\boldsymbol{I}_{smt}$

and $\boldsymbol{G}_{smt}$ are shown in Fig. 4 marked by "seg" and "smt", respectively.

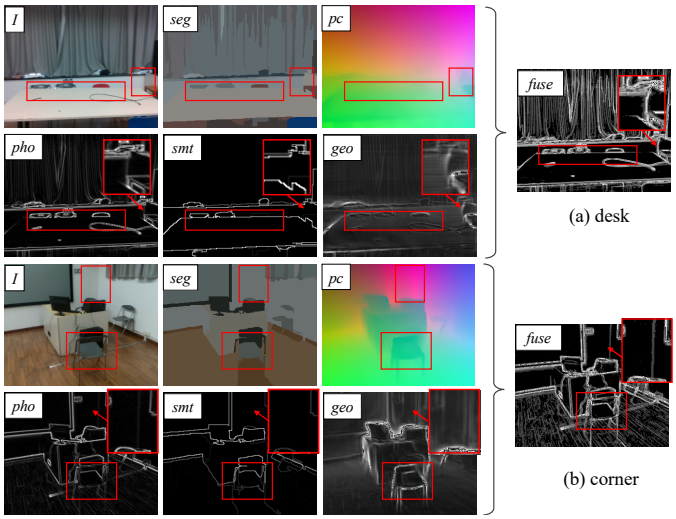

Fig. 4. Visualization of "multi-grad" maps in two typical scenes. The label "I", "seg" and "pc" denote the original RGB image, the pseudo-color maps of the segmentation result and that of the point clouds, respectively. Under them are the corresponding gradient maps, where "pho", "smt", "geo" represent the visualization results of $\boldsymbol{G}_{pho}, \boldsymbol{G}_{smt}$ and $\boldsymbol{G}_{geo}$, respectively. The final fused "multi-grad" map is marked by $fuse$.

Geometric gradient. Although semantic segmentation offers great help in distinguishing different classes of objects, it still gives confusing results when confronted with the different instances belonging to the same class. Taking the wall in Fig. 4 (b) as an example, it contains two planes perpendicular to each other, the separating line of which can not be observed by either the semantic gradient or the photometric gradient, leading to overly smooth depths in the corner. Luckily, the geometric structure can offer great help because the positions of the 3D points from the point cloud generated by the predicted pose and depths usually go through abrupt changes at the separating line of different instances, even though they belong to the same semantic class. To describe these changes, for each input image $\boldsymbol{I}_t$, we generate a pseudo colored position map denoted by $\boldsymbol{I}_{geo}$ where the 3D coordinates of all pixels are stored in the corresponding grids. Then the geometric gradient $\boldsymbol{G}_{geo}$ is acquired by $\boldsymbol{G}_{geo} = \nabla^2 \boldsymbol{I}_{geo}$. The effectiveness of $\boldsymbol{G}_{geo}$ in dealing with the challenging corner situation can be clearly seen in Fig. 4 (b) since the separating line of the walls can only be captured in $\boldsymbol{G}_{geo}$.

Gradient fusion. With the multi-source gradients $\boldsymbol{G}_{pho}, \boldsymbol{G}_{smt}$ and $\boldsymbol{G}_{geo}$, we can formulate the final "multi-grad" gradient map $\boldsymbol{G}_{fuse}$ as:

$$\boldsymbol{G}_{fuse} = (exp(\boldsymbol{G}_{geo}) + \frac{\mu_{geo}}{\mu_{smt}} \boldsymbol{G}_{smt}) * \boldsymbol{G}_{pho} \tag{11}$$

where μ_{geo} and μ_{smt} are the mean of $\boldsymbol{G}_{geo}$ and $\boldsymbol{G}_{smt}$, respectively, and $*$ is the elementwise multiplication. It can be seen from Fig. 4 that by fusing $\boldsymbol{G}_{smt}$ and $\boldsymbol{G}_{geo}$ to $\boldsymbol{G}_{fuse}$, the image gradient can be significantly enhanced with the assistance of semantic and geometric information, and more reasonable edges in the scene can be detected compared with just using the photometric gradient.

D. Loss Function

The overall loss function of our SEOVO is given as:

$$L_{total} = L_{align} + w_1 L_{smt} + w_2 L'_{sm} + w_3 L_{sc} \tag{12}$$

where L_{align} is the alignment loss, L_{smt} is the semantic loss, L'_{sm} stands for our newly designed fused smoothness loss, and L_{sc} refers to the scale consistency loss. The hyper-parameters are set to $w_1 = 0.5$, $w_2 = 0.1$ and $w_3 = 0.5$. Among all loss terms, the last one L_{sc} has been given in Eq. 5. Next, we will introduce the former three loss terms in detail.

The alignment loss. L_{align} is composed of the epipolar constrained photometric loss L'_{pho} and the point cloud consistency loss L_{pc}:

$$L_{align} = L'_{pho} + w_p L_{pc} \tag{13}$$

where w_p is set to 0.1. L'_{pho} and L_{pc} are calculated based on the two aligned poses $\boldsymbol{T}^S$ and $\boldsymbol{T}^G$ obtained in Sect. IV-A, respectively. Similar to Eq. 3, L'_{pho} penalizes the photometric inconsistency between the target image $\boldsymbol{I}_t$ and the synthesized one $\hat{\boldsymbol{I}}_t$ warped from the source view $\boldsymbol{I}_s$ based on $\boldsymbol{T}^S$ and the output depth $\boldsymbol{D}_t$. Formally,

$$\begin{aligned} L'_{pho} &= \frac{1}{\|\boldsymbol{\mathcal{V}}\|} \sum_{\boldsymbol{p}_t \in \boldsymbol{\mathcal{V}}} \left\| \boldsymbol{I}_t(\boldsymbol{p}_t) - \hat{\boldsymbol{I}}_t(\boldsymbol{p}_t) \right\|_1 \\ &= \frac{1}{\|\boldsymbol{\mathcal{V}}\|} \sum_{\boldsymbol{p}_t \in \boldsymbol{\mathcal{V}}} \left\| \boldsymbol{I}_t(\boldsymbol{p}_t) - \boldsymbol{I}_s(\frac{1}{Z_{\hat{\boldsymbol{p}}_t}} \boldsymbol{K} \boldsymbol{T}^S \boldsymbol{D}_t(\boldsymbol{p}_t) \boldsymbol{K}^{-1} \boldsymbol{p}_t) \right\|_1 \end{aligned} \tag{14}$$

where $Z_{\hat{\boldsymbol{p}}_t}$ is the depth of the 3D point warped by $\boldsymbol{T}^S$ in the adjacent camera's coordinate system and the SSIM loss is omitted here for brevity. Relying on $\boldsymbol{T}^S$ which is obtained by multiplying the epipolar pose with the global scale factor, L'_{pho} is capable of better guiding the pose regression under the epipolar geometric constraint without ruining the scale consistency of the VO system.

Apart from the scale ambiguity problem, we are also devoted to dealing with the accumulation drift in monocular VO by keeping the consistency of the global point clouds between adjacent frames. In detail, given the global poses $\boldsymbol{T}^G_k$ and $\boldsymbol{T}^G_{k+1}$ corresponding to two adjacent frames $\boldsymbol{I}_t$ and $\boldsymbol{I}_s$ at time k calculated by Eq. 9, we can project an arbitrary 2d point $\boldsymbol{p}_t/\boldsymbol{p}_s$ in $\boldsymbol{I}_t/\boldsymbol{I}_s$ to the global world coordinate system by:

$$\begin{aligned} \boldsymbol{Q}_t(\boldsymbol{p}_t) &= \boldsymbol{T}^G_k \boldsymbol{D}_t(\boldsymbol{p}_t) \boldsymbol{K}^{-1} \boldsymbol{p}_t \\ \boldsymbol{Q}_s(\boldsymbol{p}_s) &= \boldsymbol{T}^G_{k+1} \boldsymbol{D}_s(\boldsymbol{p}_s) \boldsymbol{K}^{-1} \boldsymbol{p}_s \end{aligned} \tag{15}$$

where $\boldsymbol{Q}_t$ and $\boldsymbol{Q}_s$ are the global point clouds recovered from $\boldsymbol{I}_t$ and $\boldsymbol{I}_s$, respectively. When the predicted poses and depths are accurate enough, the corresponding 3D points in $\boldsymbol{Q}_t$ and $\boldsymbol{Q}_s$ of two matched 2d points tend to be consistent. Based on this assumption, we can give the point cloud loss L_{pc} as:

$$L_{pc} = \frac{1}{\|\boldsymbol{\mathcal{V}}\|} \sum_{\boldsymbol{p}_t \in \boldsymbol{\mathcal{V}}} \|\boldsymbol{Q}_t(\boldsymbol{p}_t) - \boldsymbol{Q}_s(\boldsymbol{p}'_t)\|_2^2 \tag{16}$$

where $\boldsymbol{p}'_t$ is given in Eq. 1.

TABLE I
QUANTITATIVE COMPARISON ON DEPTH ESTIMATION WITH RELATED METHODS ON 7-SCENES DATASET

Scenes	SC-SfMLearner [16]		MonoIndoor [25]		TrianFlow [27]		SC-online18		Trian-online		OnlineVO [23]		SEOVO (ours)	
	Offline		Offline		Offline		Online		Online		Online		Online	
	AbsRel	Acc δ_1	AbsRel	Acc δ_1	AbsRel	Acc δ_1	AbsRel	Acc δ_1	AbsRel	Acc δ_1	AbsRel	Acc δ_1	AbsRel	Acc δ_1
Chess	0.103	0.880	0.097	0.888	0.114	0.817	0.098	0.882	0.107	0.867	0.101	0.878	**0.091**	**0.892**
Fire	0.089	0.916	0.077	0.939	0.107	0.874	0.080	0.919	0.091	0.907	0.079	0.912	**0.075**	**0.941**
Heads	0.124	0.862	0.106	0.889	0.173	0.755	0.099	0.912	0.102	0.908	0.132	0.884	**0.097**	**0.925**
Office	0.096	0.912	0.083	0.934	0.126	0.848	0.086	0.934	0.096	0.913	0.100	0.900	**0.080**	**0.936**
Pumpkin	0.083	0.946	**0.078**	**0.945**	0.112	0.893	0.124	0.868	0.101	0.904	0.123	0.893	0.103	0.901
RedKitchen	0.101	0.896	0.094	0.915	0.139	0.821	0.100	0.899	0.117	0.871	0.098	0.903	**0.088**	**0.917**
Stairs	0.106	0.855	0.104	0.857	0.167	0.746	0.106	0.871	0.129	0.831	0.129	0.829	**0.097**	**0.876**

The semantic loss. Weakly textured places are difficult to be well reconstructed due to the lack of photometric features. Fortunately, the sets of independent textureless planes and paired ones denoted by $\boldsymbol{\mathcal{P}}^I$ and $\boldsymbol{\mathcal{P}}^P$ respectively have been nicely extracted by our semantic extraction module discussed in Sect. IV-B with the assistance of semantics and our designed multi-grad map, enabling us to improve the reconstruction quality in these areas by constraining them with the semantic loss L_{smt}. First, we propose a planar loss L_{pl} as:

$$L_{pl} = \sum_{P^I \in \boldsymbol{\mathcal{P}}^I} \left\| var(\boldsymbol{\mathcal{N}}(P^I)) \right\|_2^2 + \frac{1}{N^I} \sum_{P^I \in \boldsymbol{\mathcal{P}}^I} \sum_{\boldsymbol{p}^I \in P^I} \nabla \boldsymbol{D}_t(\boldsymbol{p}^I) \tag{17}$$

where $\boldsymbol{\mathcal{N}}(P^I)$ returns the normal map of P^I which records the normal vectors of all points in it, $var(\cdot)$ calculates the channel-wise variances of the given normal map, and N^I is the number of pixels in $\boldsymbol{\mathcal{P}}^I$.

Besides, in order to cope with challenging "corner" situation such as in Fig. 4 (b), we introduce the orthogonal loss L_{org} to constrain the paired planes in $\boldsymbol{\mathcal{P}}^P$ to be perpendicular to each other:

$$L_{org} = \sum_{[P_1, P_2] \in \boldsymbol{\mathcal{P}}^P} \| mean(\boldsymbol{\mathcal{N}}(P_1)) \cdot mean(\boldsymbol{\mathcal{N}}(P_2)) \|_2^2 \tag{18}$$

where $mean(\cdot)$ calculate the channel-wise average of the given normal map.

Finally, the semantic loss can be given as:

$$L_{smt} = L_{pl} + L_{org}. \tag{19}$$

The fused smoothness loss. Different with the commonly used smoothness loss term L_{sm} introduced in Eq. 4, our fused smoothness loss L'_{sm} better guides the optimization of depths relying on our multi-grad map $\boldsymbol{G}_{fuse}$ instead of $\boldsymbol{G}_{pho}$, which introduces richer edge information consistent with perception. Specifically, L'_{sm} is given as:

$$L'_{sm} = \frac{1}{\|\boldsymbol{\mathcal{V}}\|} \sum_{\boldsymbol{p}_t \in \boldsymbol{\mathcal{V}}} (exp(-\boldsymbol{G}_{fuse}(\boldsymbol{p}_t)) \cdot \nabla \boldsymbol{D}_t(\boldsymbol{p}_t))^2. \tag{20}$$

V. EXPERIMENTAL RESULTS

A. Experimental Setup

Implementation details. All experiments in this paper were conducted on the same desktop computer equipped with the GPU of NVIDIA GeForce RTX 3070. The training and evaluation codes of SEOVO were implemented using PyTorch. As for network architectures, we borrowed Flownet [27] and used the same PoseNet and DepthNet as [16]. The self-supervised semantic segmentation network proposed in [41] was employed as SegNet. All images were resized to 320×256 and then fed to SEOVO in a time order. The input frame will be selected as a new keyframe if the mean flow of correspondences between it and the last keyframe was larger than 20, and only keyframes will be used in regression so as to remove redundant frames. The weights of the network model were initialized by the pretrained weights on the NYUv2 dataset [42]. During the online training, the Adam optimizer was used with 10^{-4} as the initial learning rate.

Datasets. First, we evaluated our proposed SEOVO on a public indoor dataset, the RGB-D 7-scenes dataset [43] which is collected in seven typical indoor scenes. Then to better confirm the benefits of our SEOVO in textureless environments, we also collected two video sequences by a Realsense d453i camera in our lab and a closed corridor namely "Lab" and "Corridor", respectively, for online testing. These sequences include large areas of texturelss regions which are difficult to be handled by existing deep VO schemes. Both the public dataset and our collected one provide RGB sequences and the depth maps corresponding to each frame utilized as the ground truths. Besides, the 7-scenes dataset provides the frame-to-frame poses for evaluating the accuracy of pose regression. It is worth mentioning that, in our experiments, the offline schemes were trained on the training data of the evaluation dataset before inference while the online ones were all initialized by the pretrianed model on NYUv2 and were directly optimized on the testing data to make predictions.

Evaluation metrics. The evaluation of SEOVO mainly focused on two aspects, the performance on the depth estimation and the pose estimation. In terms of the depth evaluation, we followed [16] to use the mean absolute relative error (AbsRel), the root mean square error (RMS), and the accuracy under threshold ($\sigma_i \leq 1.25^i, i = 1, 2, 3$) as metrics. On the other side, to measure the accuracy of the output pose, we selected two widely used metrics in measuring the SLAM trajectory, the Absolute Pose Error (APE) and the Relative Pose Error (RPE), which investigate the global and the local consistency of the trajectory, respectively. Specifically, we calculated RMS of the above two pose metrics on the translation and the rotation components, represented by APE_t, APE_r, RPE_t, and RPE_r. The subscript "t" corresponds to the translation component while "r" indicates the rotation one.

B. Quantitative Experiment

Depth estimation on 7-scenes. We first presented the quantitative depth estimation results of our SEOVO and several offline or online competitors on the 7-scenes dataset in Table I. Considering that there are few public online VO methods, we extended offline works [15], [27] which are closely related to our SEOVO to the online versions for comparison, denoted by SC-online18 and Trian-online, respectively. Technically, these extended online schemes were optimized based on the same network architectures and loss functions as their original works, but under an online framework. From the results presented in Table I, we make the following observations. First, both the offline SC-SfMLearner and TrianFlow underperform their online variants SC-online18 and Trian-online, respectively, proving the superiority of online learning in the generalization ability. Second, the offline MonoIndoor gives satisfactory results, even outperforming most online competitors since it is the only counterpart specially designed for indoor environments, which improves the performance of the depth estimation by predicting the indoor depth range. This phenomenon proves our claim that indoor environments are challenging and should be carefully dealt with. Third, thanks to our designed multi-grad map and the semantic extraction module, SEOVO shows an overwhelming performance in nearly all scenes, which integrates the geometric structure and the semantic layout of the scene to the online optimization framework, thus improving the accuracy of the depth estimation in indoor environments.

TABLE II
QUANTITATIVE COMPARISON WITH ONLINE METHODS ON OUR COLLECTED DATASET

Scene	Method	AbsRel	RMS	σ_1	σ_2	σ_3
Lab	SC-online18 [15]	0.118	0.720	0.872	0.991	0.995
	Trian-online [27]	0.155	0.955	0.786	0.948	0.986
	OnlineVO [23]	0.119	0.765	0.867	0.992	0.997
	SEOVO(ours)	**0.113**	**0.708**	**0.893**	**0.996**	**0.999**
Corridor	SC-online18 [15]	0.139	0.776	0.837	0.970	0.989
	Trian-online [27]	0.176	0.958	0.808	0.937	0.969
	OnlineVO [23]	0.146	0.815	0.827	0.967	0.988
	SEOVO(ours)	**0.137**	**0.752**	**0.844**	**0.972**	**0.990**

Depth estimation on our dataset. To demonstrate the performance of SEOVO in tackling the challenging environment with weak textures, we also conducted experiments on our collected testing data containing large areas of textureless regions. To directly evaluated on our testing sequences, we compared our SEOVO with online schemes free of finetune training and gave their evaluation results in Table II. It can be seen from Table II that our scheme exhibits clear performance advantages over all counterparts in this challenging dataset owning to the semantic loss which constrains the depth regression in the textureless regions, further proving the effectiveness of semantic assistance.

Pose estimation. To evaluate the accuracy of our visual odometry in the pose estimation, we aligned the output trajectories predicted by our SEOVO and typical competitors to the ground truth one respectively and measured the aligned results based on the four metrics mentioned in Sect. V-A, whose results were reported in Table III. Whether from the absolute metrics or the relative ones in Table III, our SEOVO improves the accuracy of the estimated poses. The underlying reason is that our epipolar constrained photometric loss guides SEOVO to optimize the estimated pose under the epipolar geometric constraint and to maintain both the photometric and geometric consistency, which undoubtedly ensuring an excellent pose prediction. Also, SEOVO reduces the accumulation drift in the whole trajectory by the global optimization on the point clouds, further improving the localization accuracy.

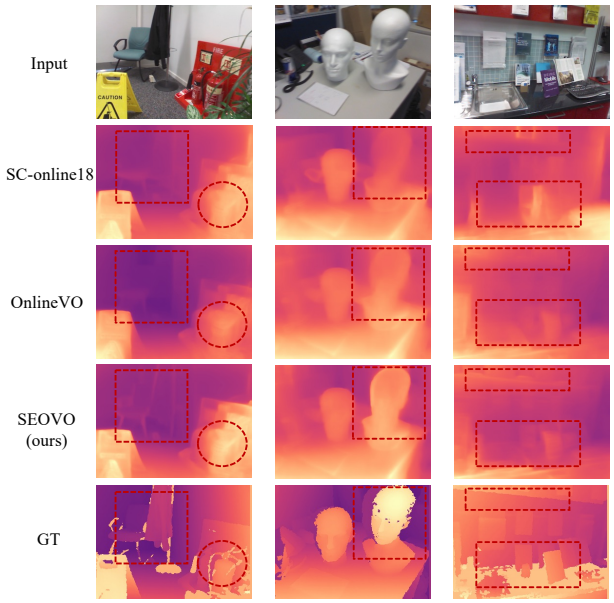

Fig. 5. Depth estimation results on the 7-scenes dataset. Thanks to our two innovated modules introducing the semantic and geometric information, SEOVO performs favorably compared to other competitors in better prediction results and preserves sharper edges. The details deserved to be noticed are inside the bounding boxed with red dotted lines.

C. Qualitative Experiment

Visualized depth maps. In order to qualitatively examine the performance of SEOVO, we visualized the depth estimation results on 7-scenes of our SEOVO and other competitors in Fig. 5. Besides, we also provided the ground truth depths denoted by "GT" as reference. It can be clearly seen from Fig. 5 that the depth maps generated by our SEOVO are closer to the ground truth maps, which corroborates to our quantitative results in Table I indicating our remarkable performance in depth estimation. Besides, our depth maps have sharper edges thanks to integrating our designed multi-grad map introduced in Sect. IV-C, which is contributed to guiding the optimization of depth smoothness by edges detected in this fused gradient map under the photometric, the geometric and the semantic constraints.

Reconstruction results. To show the performance of our SEOVO in both the depth estimation and the pose regression qualitatively, we offered some typical results of the point clouds synthesized by the poses and depths output from SC-online18, OnlineVO and our SEOVO in Fig. 6. In detail, for the completeness of maps, we transformed the synthesized point clouds from several consecutive frames (usually 3 to

TABLE III
QUANTITATIVE COMPARISON ON POSE ESTIMATION WITH RELATED METHODS ON 7-SCENES DATASET

Scenes	SC-SfMLearner [15]				SC-online18 [15]				Trian-online [23]				OnlineVO [23]				SEOVO (ours)			
	ATE_t	ATE_r	RTE_t	RTE_r	ATE_t	ATE_r	RTE_t	RTE_r	ATE_t	ATE_r	RTE_t	RTE_r	ATE_t	ATE_r	RTE_t	RTE_r	ATE_t	ATE_r	RTE_t	RTE_r
Chess	0.416	11.774	0.073	2.256	0.318	10.457	0.053	1.687	0.443	15.871	0.076	2.203	0.278	4.678	0.043	1.311	**0.246**	**4.385**	**0.040**	**1.211**
Fire	0.496	29.478	0.094	3.216	0.397	24.039	0.054	2.916	0.460	60.016	0.086	2.622	0.338	25.191	0.061	2.436	**0.284**	**23.111**	**0.054**	**2.087**
Heads	0.176	26.431	0.034	2.282	0.148	23.482	0.025	1.623	0.166	17.384	0.040	2.153	0.144	16.944	0.024	1.450	**0.125**	**13.567**	**0.024**	**1.370**
Office	0.311	33.455	0.042	1.943	0.189	25.717	0.035	1.621	0.435	31.052	0.062	2.327	0.190	26.255	0.039	1.984	**0.151**	**21.483**	**0.033**	**1.582**
Pumpkin	0.379	23.744	0.055	1.578	0.244	7.840	0.075	1.143	0.495	38.235	0.077	1.744	0.204	10.26	0.071	1.134	**0.185**	**7.437**	**0.067**	**1.085**
RedKitchen	0.229	18.878	0.045	2.660	1.074	13.025	0.040	1.085	0.263	36.75	0.060	1.359	0.178	12.431	0.039	1.075	**0.169**	**10.758**	**0.038**	**1.074**
Stairs	0.533	159.473	0.282	2.660	0.410	101.35	0.219	2.067	0.756	29.731	0.265	2.867	0.394	57.758	0.210	2.429	**0.303**	**22.852**	**0.172**	**1.945**

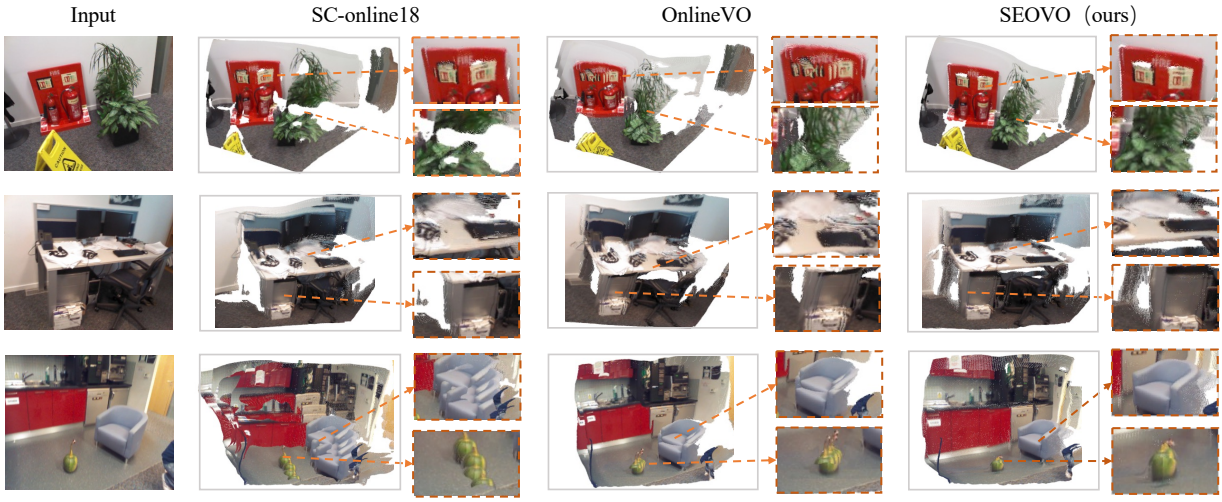

Fig. 6. Illustration of the point clouds synthesized by the poses and depths generated by the online baseline SC-online18, OnlineVO and our SEOVO on 7-scenes. To show the global consistency of the estimation, we warp the maps of several consecutive keyframes to a unified global coordinate system for more complete model.

5 keyframes) to a unified global coordinate system which takes the first frame as reference. It can be seen from Fig. 6 that the results of other two competitors have obvious misalignments among frames while our SEOVO best keeps the consistency of global maps, corroborating to the claim that our scheme produces the most accurate depths and poses. The most important reason explaining for our SEOVO's superiority in reconstruction quality is that our SEOVO directly penalizing the distance of the point clouds between frames while other competitors only constrain the depth consistency. Moreover, the reconstruction results of the baseline SC-online18 and our SEOVO in the challenging textureless environments were given in the second and the last column in Fig. 7, respectively. It can be seen that the walls or the blackboard reconstructed by SC-online18 are twisted owing to the overfitting of the photometric loss in these regions while our SEOVO does a great job in recovering the geometric structure of the texturelss regions by extracting these planes with the assistance of semantics and impose planar constraints on them.

TABLE IV
PERFORMANCE OF NETWORKS TRAINED WITH VARIOUS COMBINATIONS OF LOSS TERMS ON 7-SCENES AND OUR COLLECTED DATASET

Scenes	EOVO		SOVO		SEEOVO		SEOVO	
	AbsRel	Accδ_1	AbsRel	Accδ_1	AbsRel	Accδ_1	AbsRel	Accδ_1
Chess	0.097	0.883	0.097	0.882	0.104	0.876	**0.091**	**0.892**
Fire	0.087	0.911	0.086	0.917	0.090	0.903	**0.075**	**0.925**
Heads	0.100	0.913	0.101	0.916	0.103	0.901	**0.097**	**0.925**
Office	0.087	0.930	0.092	0.926	0.091	0.920	**0.080**	**0.936**
Pumpkin	0.128	0.873	0.128	0.873	0.132	0.831	**0.103**	**0.901**
RedKitchen	0.095	0.910	0.108	0.885	0.118	0.872	**0.088**	**0.911**
Stairs	0.102	**0.888**	0.115	0.857	0.134	0.833	**0.097**	0.876
Lab	0.120	0.861	0.121	0.860	0.122	0.870	**0.113**	**0.893**
Corridor	0.142	0.832	0.158	0.802	0.147	0.821	**0.137**	**0.844**

TABLE V
QUANTITATIVE COMPARISON WITH DIFFERENT VARIANTS ON POSE ESTIMATION

Scenes	EOVO		SOVO		SEEOVO		SEOVO	
	RTE_t/m	$\mathrm{RTE}_r/°$	RTE_t/m	$\mathrm{RTE}_r/°$	RTE_t/m	$\mathrm{RTE}_r/°$	RTE_t/m	$\mathrm{RTE}_r/°$
Chess	0.047	1.437	0.501	1.525	0.068	2.020	**0.040**	**1.211**
Fire	0.074	2.627	0.080	2.549	0.072	2.485	**0.054**	**2.087**
Heads	0.026	1.699	0.026	1.542	0.028	1.618	**0.024**	**1.370**
Office	0.033	1.586	0.037	1.734	0.041	1.856	**0.033**	**1.582**
Pumpkin	0.074	1.152	0.072	1.137	0.092	1.398	**0.067**	**1.085**
RedKitchen	0.040	1.095	0.043	1.083	0.041	1.167	**0.038**	**1.074**
Stairs	0.239	**1.793**	0.195	1.868	**0.109**	1.990	0.172	1.945

D. Ablation Study of Loss Terms

We demonstrated how two important loss terms, the alignment loss and the semantic loss in our loss function affect the results by comparing SEOVO with its two variants EOVO and SOVO. Specifically, compared with SEOVO, EOVO is optimized without the semantic loss L_{smt} while SOVO is without the alignment loss L_{align}. Furthermore, to verify the effectiveness of implicitly introducing the geometric constraint to SEOVO as the epipolar constrained photometric loss L'_{pho}, we added an competitor named SEEOVO which replaces L'_{pho} with the direct pose supervision $L_{exp} = \left\| \boldsymbol{T}^N - \boldsymbol{T}^E \right\|_2^2$ where $\boldsymbol{T}^E$ is the epipolar pose and $\boldsymbol{T}^N$ is the normalization result

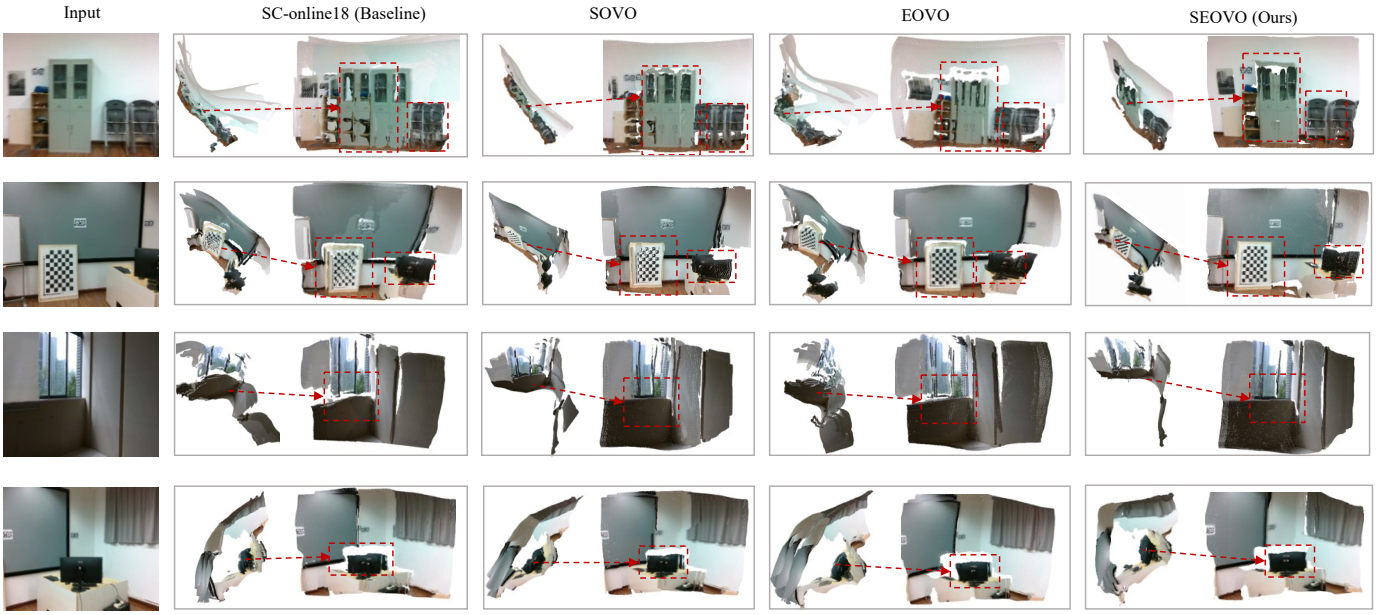

Fig. 7. Comparisons of the reconstructed maps on our collected dataset with online baseline SC-online18, and our two variants. To clearly demonstrate the reconstruction results in the textureless regions, we provide both the top views and the font views in each block.

of the local predicted pose $\boldsymbol{T}^L$.

Table IV and Table V tabulate the quantitative results of SEOVO and its rivals on the depth estimation and the pose estimation, respectively. These two tables demonstrate that our SEOVO outperforms other variants in both depth and pose regression except in the "Stairs" scene. The reason of our unsatisfactory performance in this scene lies in the strong reflection of sunshine on the stairs, which coincides with our failure cases with complex reflection conditions discussed in Sect. V-E. Technically, EOVO imposes the geometric constraints on the pose regression while SOVO guarantees the performance in textureless regions under planar constraints with semantic assistance. Compared with them, SEOVO combines both geometric and semantic cues, undoubtedly achieving the best performance. Besides, there's an interesting finding that SEEOVO performs worst in most scenes, which validates our analysis in Sect. IV-A that utilizing the explicit pose constraint limits the VO performance due to the neglect of the scale information.

In addition to the quantitative comparison, Fig. 7 qualitatively shows the point clouds synthesized by the poses and depths from multiple frames estimated by our SEOVO and its variants SOVO and EOVO. For clearer observations, we provided both the top and the front views of them. It can be clearly seen from the top view in Fig. 7 that SOVO achieves remarkable results in reconstructing the geometric structures of low-texture regions only inferior to our SEOVO, proving the effectiveness of the semantic loss which enforces the extracted surface points to be on the same plane and constrains the walls in the corner to be perpendicular to each other. However, it fails to output poses with high accuracy to warp the 3D points to the global coordinate system, thus generating obvious misalignments among frames clearly shown in the front view. On the other hand, although EOVO guarantees the pose accuracy by the epipolar geometric constraint introduced in L'_{pho}, it still highly relies on the photometric consistency to guide the depth optimization, which is unreliable in low-texture regions, thus causing the poor reconstruction quality in these places. Fortunately, our SEOVO integrates both the geometric cues and the semantic ones, generating the best reconstruction results observed from both the front view and the top view. These results lead us to express the belief that the loss function of SEOVO is well designed and both the alignment loss and the semantic loss play essential roles in it.

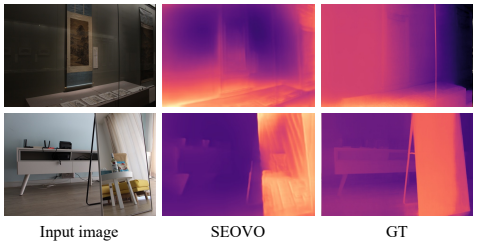

Fig. 8. Depth estimation of failure case with the complex reflection owning to the glass and the mirror. In detail, the glass part is hard to distinguish and recover the true depth while the artifacts in the mirror is mistaken for the objects with real depth.

E. Failure Case Analysis

By observing and analyzing the experimental results, we found that the illumination and reflection of the scene have an obvious influence on SEOVO's performance. For example, when an object's surface is made of strongly reflective materials such as glass or mirror, information contained in the imaging areas of these objects is relatively confusing. In these cases with complex reflections, the basic assumption

of the photometric consistency is violated and the impact of noises will be more notable. This problem poses a challenge to SEOVO in extracting high-quality features from deep levels and the correspondences generated by FlowNet tend to be mismatched, resulting in poor performance shown in Fig. 8. It can be seen from the figure that the glass window in front of the artwork is hard to be reconstructed while the artifacts in the mirror cause the pseudo depths.

VI. Conclusion And Feature Work

In this paper, we studied a practical problem, the indoor deep visual odomertry which is of great significance in visual perception tasks such as AR, indoor navigation and smart homes. Considering the limitations of existing methods in accuracy and generalization ability, we proposed a fully self-supervised solution under an online adaptive framework namely SEOVO tailored to tackle two challenging problems of self-supervised VO in indoor environments. These problems are the difficulty in regressing the complex indoor motions and in handling the low-texture environments. First, we design an epipolar constrained photometric loss to implicitly impose the geometric constraint on the pose regression, which not only improves the estimation accuracy but also nicely deals with the scale ambiguity of monocular VO. Second, the "Semantic Extraction Module" was proposed to finely extract the low-texture planes with the help of our designed "multi-grad" map and these planes are then addressed with planar constraints, which guarantees the reconstruction quality with the support of semantic prior. What's more, the "multi-grad" map combining multi-source gradients induced by the photometric, the geometric and the semantic constraints contributes to generate better edge-aware depth maps. Capitalizing on the above innovations, our SEOVO shows remarkable performance not only in the public datasets but also in our collected low-texture dataset. Though SEOVO can work well in most cases, its performance is still not satisfied when working in environments with complex reflections. In our future work, we will continue to devote our efforts in this area.

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